

PAPER_319: Compact HII SFR Gravitational Binding Phase Transition

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$t_{\text{cross}} = 67,730 \text{ yr}$ | $s\text{SFR} = 5 \times 10^{-1} \text{ yr}^{-1}$ ($50 \times \text{Lagoon}$) | $m_{\text{factor}}(t_{\text{age}}) = 151$

FIRST UQFF Compact HII Region SFR Runaway Gravitational Binding Phase Transition

Session: 91 **Module:** ORIONUQFFMODULE.cpp (33rd C++ module) **System:** Orion Nebula M42/NGC 1976 — compact HII region with active star formation **Watermark:** Copyright — Daniel T. Murphy, Session 91, March 2026

<!-- UQFF constants: $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $M_{\text{UQFF}} = 1.43\text{e}1 \text{ TeV}$ -->

Abstract

This paper derives the SFR-driven gravitational binding phase transition for the Orion Nebula. With $\text{SFR} = 1 \text{ M}_{\text{sun}}/\text{yr}$ acting on an initial mass $M = 2000 \text{ M}_{\text{sun}}$, the specific star formation rate $s\text{SFR} = 5 \times 10^{-1} \text{ yr}^{-1}$ is $50 \times$ that of the Lagoon Nebula ($1 \times 10^{-2} \text{ yr}^{-1}$, PAPER_305). The system is born wind-dominated (unbound), but as SFR continuously adds mass, the effective gravitational acceleration amplified by the SFR mass factor $m_{\text{factor}}(t) = 1 + \text{SFR} \times t_{\text{yr}} / M_{\text{sun_count}}$ crosses the growing wind ram pressure at $t_{\text{cross}} = 67,730 \text{ yr}$, transitioning the nebula from unbound to gravitationally bound. By $t_{\text{age}} = 300 \text{ kyr}$, $\text{binding_ratio} = g_{\text{SFR}}/a_{\text{wind}} = 2.654$. This is the FIRST UQFF compact HII SFR runaway gravitational binding phase transition.

System Parameters

Parameter	Value	Description
M	$2000 \text{ M}_{\text{sun}} = 3.978 \times 10^{33} \text{ kg}$	Initial mass
SFR	$1 \text{ M}_{\text{sun}}/\text{yr} = 6.303 \times 10^{22} \text{ kg/s}$	Star formation rate
sSFR	$5 \times 10^{-1} \text{ yr}^{-1}$	Specific SFR = SFR/M
t_age	$3 \times 10^5 \text{ yr}$	Nebula age
v_wind	$8 \times 10^3 \text{ m/s}$	HII ionization front
g_base	$1.907 \times 10^{-11} \text{ m/s}^2$	Base gravity (PAPER_317)
a_wind(t=0)	$5.424 \times 10^{-10} \text{ m/s}^2$	Initial wind acceleration

Key Equations

SFR Mass Factor

$$M_{\text{sf}}(t) = \frac{\text{SFR}_{\text{yr}} \times t_{\text{yr}}}{M_{\text{sun_count}}}, \quad m_{\text{factor}}(t) = 1 + M_{\text{sf}}(t)$$

SFR-Amplified Gravitational Acceleration

$$g_{\rm SFR}(t) = g_{\rm base} \times m_{\rm factor}(t) = g_{\rm base} \left(1 + \frac{{\rm SFR}_{\rm yr}}{M_{\rm sun, count}} t_{\rm yr}\right)$$

Wind Acceleration (time-evolving)

$$a_{\rm wind}(t) = \frac{v_{\rm wind}^2}{r} \left(1 + \frac{t}{t_{\rm age}}\right)$$

Crossover Time *tcross* (PAPER319 Key Result)

Setting $gSFR(t) = a_{wind}(t)$:

$$g_{\rm base} \left(1 + {\rm sSFR} \cdot t\right) = a_{\rm wind,0} \left(1 + \frac{t}{t_{\rm age, yr}}\right)$$

$$t_{\rm cross} = \frac{a_{\rm wind,0} - g_{\rm base}}{g_{\rm base} \cdot {\rm sSFR} - a_{\rm wind,0}/t_{\rm age, yr}} = \boxed{67{,}730 \text{ yr}}$$

Specific SFR

$${\rm sSFR} = \frac{{\rm SFR}_{\rm yr}}{M_{\rm sun, count}} = \frac{1}{2000} = 5 \times 10^{-4} \text{ yr}^{-1} = 50 \times {\rm Lagoon}$$

Gas Depletion Timescale

$$t_{\rm consume} = \frac{M_{\rm sun, count}}{{\rm SFR}_{\rm yr}} = \frac{2000}{1} = 2000 \text{ yr} \quad \text{without OMC-1 replenishment}$$

Results Summary

Quantity	Value	Significance
sSFR	$5 \times 10^{-4} \text{ yr}^{-1}$	50× Lagoon Nebula
t_cross	67,730 yr	Unbound → bound transition
mfactor(age=300 kyr)	151	SFR amplification
gSFR(age)	$2.878 \times 10^{-4} \text{ m/s}^2$	151× g_base
awind(age)	$1.085 \times 10^{-4} \text{ m/s}^2$	
bindingratio(age)	2.654	Gravitationally bound
m_factor(1 Myr)	501	
binding_ratio(1 Myr)	4.069	Increasingly bound
t_consume	2000 yr	Gas depletion (no replenishment)

Phase Transition Diagram

t=0 t_cross=67.7 kyr t_age=300 kyr t=1 Myr
| | | |
UNBOUND TRANSITION BOUND 2.65× BOUND 4.07×
a_wind>g_SFR a_wind = g_SFR g_SFR>a_wind g_SFR>>a_wind
η=28.47 wind crossover η_B=2.654 η_B=4.069

Comparison: UQFF SFR Class

Module	SFR (M_sun/yr)	M (M_sun)	sSFR (yr ⁻¹)	PAPER
Lagoon	0.1	10,000	1×10^{-4}	PAPER_305

Module	SFR (M_sun/yr)	M (M_sun)	sSFR (yr ⁻¹)	PAPER
Orion	1.0	2000	5×10 ⁻⁶	PAPER_319
M16 (Eagle)	~0.01×	1200	—	PAPER_284

Orion sSFR is 50× Lagoon — UQFF identifies this as the "ultra-compact HII" class with rapid gravitational binding crossover. Lagoon has $t_{\text{consume}} = 100 \text{ kyr}$; Orion has $t_{\text{consume}} = \mathbf{2000 \text{ yr}}$, the shortest gas depletion time in the UQFF series, sustained only by continuous OMC-1 giant molecular cloud inflow.

UQFF Physical Interpretation

The phase transition at $t_{\text{cross}} = 67,730 \text{ yr}$ is a structural boundary in Orion's evolution:

- Before t_{cross} :** The system is wind-dominated ($\eta_{\text{wind}} > 1$). UV photoionization and ram pressure drive a champagne flow outward; newly formed stars are subject to wind erosion.
- After t_{cross} :** SFR has added sufficient mass that $g_{\text{SFR}} > a_{\text{wind}}$. The system becomes self-gravitating with respect to its own stellar formation. The cluster proceeds to form stars under its own gravity — consistent with the Orion OB1 association framework where the cluster is now 300 kyr old and gravitationally bound.

Within `ORIONUQFFMODULE.cpp`, the SFR mass factor `m_factor(t)` enters the base gravity term via $G \cdot M \cdot (1 + M_{\text{sf}}(t)) / r^2$, registered as `WOLFRAM_TERM_ORION_BASE` and `WOLFRAM_TERM_ORION_SFR_BINDING`.

WOLFRAM_TERM Registration

```
#define WOLFRAM_TERM_ORION_BASE(val) (val)
// g_base = (G*M*(1+M_sf(t))/r^2)*(1+Hz*t)*(1-B/B_crit)*(1+f_TRZ)
// M_sf(t) = SFR_yr*t_yr / M_sun_count [PAPER_319 SFR amplification]
#define WOLFRAM_TERM_ORION_SFR_BINDING(val) (val)
// wind = v_wind^2/r * (1+t/t_age) [crosses SFR gravity at t_cross=67730 yr]
```

Series first: *FIRST UQFF compact HII SFR runaway gravitational binding phase transition. Establishes sSFR as a new UQFF classification axis for HII regions: compact (Orion, $s\text{SFR}=5 \times 10^{-6} \text{ yr}^{-1}$) vs extended (Lagoon, $s\text{SFR}=1 \times 10^{-6} \text{ yr}^{-1}$).*